

Requirement Analysis Document (RAD)

Project Title: OmniCart: Multi-Vendor E-Commerce Platform (Simple Version)

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1. Executive Project Summary

OmniCart is an online shopping platform where many different vendors can sell their products, similar to Amazon or Flipkart. Customers can sign up, browse products, place orders, pay securely, and track their deliveries in real time.

Running a large online store safely needs a strong, reliable database. For example, when a customer buys the last item in stock, the stock count must go down right away so no one else can accidentally buy the same item. The payment system must confirm the payment, and the delivery system must get the order details instantly.

To keep everything correct, OmniCart uses a relational database (like PostgreSQL or Oracle) and follows the **ACID** (Atomicity, Consistency, Isolation, Durability) rules. For example:

- **Atomicity (A):** When an order is placed, the database must do two things together: mark the payment as Successful and reduce the item's stock. If the payment fails partway through, everything is cancelled — the stock stays the same, and the customer is not charged.
- **Consistency (C):** The database only allows valid data. A product price can never be saved as a negative number ($\text{Price} \geq 0$), and an order item can never point to a product that doesn't exist.

2. Core Business Domains & Features

Based on how the store works, the database requirements are grouped into four main modules:

2.1 User Registration & Account Management

This module handles customer sign-up, profiles, login security, and loyalty levels.

- **Business Rules:**
 - Every user must sign up with a unique email address. This email is used as the login ID, so no two accounts can share one.
 - When someone signs up, the system gives them a unique User ID and stores their Name, Phone Number, Shipping Address, and the date they joined (this date never changes).
 - Customers are placed into a "Loyalty Tier" (Standard, Premium, VIP) based on how much they've spent over time.
 - A user profile can exist in the system even if they haven't bought anything yet.

2.2 Product Catalog & Browsing Engine

This module manages the product listings from all vendors, along with categories and search.

- **Business Rules:**

- Every product must belong to at least one category (e.g., "Electronics", "Apparel").
- Vendors provide product details: Title, Brand, Description, Images, and a unique SKU code.
- The system tracks live stock for every item and blocks checkout automatically once stock reaches zero.

2.3 Order Placement & Lifecycle Engine

This is the core of the system — it handles the shopping cart, checkout, and order tracking.

- **Business Rules:**

- Every checkout creates one Order ID, linked to exactly one Customer.
- One order can hold many products in different amounts. This is a many-to-many relationship (M:N) between Orders and Products, managed with a bridge table called Order Details.
- **The Historic Price Rule:** The price stored in Order Details must be exactly what the price was at the moment of purchase. If a vendor changes the price later, it must not change past orders — this keeps accounting records accurate.
- Every order must follow this fixed journey: Cart Created → Pending Payment → Paid/Confirmed → Shipped → Delivered.

2.4 Payments & Shipment Tracking Operations

This module manages payments, currency checks, and delivery tracking.

- **Business Rules:**

- Each order is linked to exactly one Payment Profile (a 1:1 relationship).
- An order cannot move to "Confirmed" until the payment gateway confirms the payment was Successful.
- Once payment is confirmed, the system creates a unique Tracking ID. The order status changes to "Shipped," then to "Delivered" once the courier's system confirms it reached the customer's address.

3. Actor Profiles & System Interactions

To keep the system secure, different types of users are given different access levels. There are four main actors:

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| Actor | --> | Read Privileges | --> | Write/Edit Rules |
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| Customer | | Catalogs, Reviews | | Register, Place Orders|
| Vendor   | | Sales Reports   | | Update Prices/Stock   |
| Delivery Agent| | Order Addresses | | Update Shipping Status|
| System Admin | | Full DB Tables | | Global Database CRUD |
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1. Customer

- **Can Read:** Product catalogs, reviews, estimated delivery times, and their own order history.
- **Can Write/Edit:** Profile details, cart changes, checkout, payments, and product reviews.
- **Safety Boundary:** Cannot see other users' addresses or payment details, or a vendor's sales numbers. Cannot change product prices.

2. Vendor (Merchant)

- **Can Read:** Their own product stock, customer reviews, and their own sales reports.
- **Can Write/Edit:** Add new products, update stock levels, change prices, and set discounts.
- **Safety Boundary:** Cannot touch payment records, see other vendors' data, or access global system settings.

3. Delivery Agent (Logistics Partner)

- **Can Read:** Customer name, delivery address, phone number, and package details.
- **Can Write/Edit:** Update the delivery status from "Shipped" to "Delivered."
- **Safety Boundary:** Cannot view payment card details, change order quantities, or edit product prices.

4. System Admin

- **Can Read/Write:** Full control (Create, Read, Update, Delete) over every table in the database.
- Every action an Admin takes is automatically saved in a secure audit log, so nothing can be changed without a record.

4. Analytical Reporting Requirements

To help the store run efficiently, the database supports three reports built from multiple tables:

Report 1: Low-Stock Product Alerts

- **Business Value:** Helps prevent popular products from running out, so restocking can happen automatically.
- **What it shows:** Vendor Name, Product SKU, Product Title, Current Stock, and how far below the safety limit the stock has dropped — for every low-stock item.

Report 2: Top-Tier VIP Customer Report

- **Business Value:** Helps find high-value customers for special offers, exclusive deals, and priority support.
- **What it shows:** Customer Name, Email Address, number of orders, and total amount spent over the last 12 months, sorted from the highest spender down.

Report 3: Vendor Category Sales Directory

- **Business Value:** Shows which product categories earn the most money across all vendors, helping plan future promotions.
- **What it shows:** For a chosen month — total sales value, number of items sold, and average customer rating, grouped by category.

5. Non-Functional Requirements & Database Constraints

These rules keep the database secure, fast, and reliable in real-world use:

Strict Data Constraints

- Review ratings must be whole numbers from 1 to 5 ($1 \leq \text{Rating} \leq 5$).
- Product prices and shipping charges must always be greater than zero ($\text{Price} > 0$).
- No two active deliveries can share the same Tracking ID — the database enforces this with a unique key.

Concurrency Controls

- If two customers try to buy the very last item at the exact same moment, the database temporarily locks that product's record (a Row-Level Transaction Lock). The first buyer's payment is processed and stock drops to zero, then the lock is released. The second buyer's system then sees the item is out of stock and is told politely, instead of causing an error or a double sale.

Historical Auditing & Safe Deletion

- User accounts are never permanently deleted. If someone closes their account, their record is simply marked "Inactive," and personal details like phone number and address are hidden or cleared for privacy. Their past orders, order numbers, and payment records stay in the database so that yearly financial audits stay accurate and complete.